1. Role of MHC, diff b/w MHC I and II

These transport molecules are called the Major Histocompatibility Complex (**MHC**) proteins. Without these, there would be no presentation of internal or external antigens to the T cells. The **importance of MHC** proteins is that they allow T cells to distinguish self from non-self.

MHC Class I The side chains of other amino acids of the peptide point outward, in a position to be recognized by receptors on cytotoxic T cells.

MHC Class II Class II can accommodate longer peptides, which are usually 13–17 amino acids. Peptide not bound by ends. It is held in the groove by parts of its peptide backbone that bind to invariant pockets formed by conserved amino acids that line all class II MHC peptide-binding grooves

2. CAm kinase function

CaM kinaseII or CaMKII is a serine/threonine-specific protein**kinase** that is regulated by the Ca^{2+} /calmodulin complex. CaMKII is involved in many signaling cascades and is thought to be an important mediator of learning and memory.

3. Homtypic heterotypic fusion

Heterotypic fusion occurs between compartments that differ in composition. For example, a vesicle derived from one compartment fuses heterotypically with a different compartment.

Homotypic fusion occurs between compartments that are biochemically identical. For example, endosomes fuse with one another, as do lysosomes, the ER and the mitochondria.

4. Confluent monolayer & past confluent

Confluent monolayer refers to cells in tissue culture, e.g. epithelial cells, which form a cohesive sheet comprising of a single cell layer filling the entire surface area of the bottom of the culture dish. Cells have then reached **confluence**.

5. Banzer exp result

The **T4 *r*II system** is an experimental system developed in the 1950s by Seymour Benzer for studying the substructure of the gene. The experimental system is based on genetic crosses of different mutant strains of bacteriophage T4, a virus that infects the bacteria *E. coli*.

6. 5 types of antibiotic

IgM, IgD, IgG, IgE, IgA

7. Restriction endonuclease

an enzyme produced chiefly by certain bacteria, that has the property of cleaving DNA molecules at or near a specific sequence of bases.

8. Pcr requirements

The various components **required** for **PCR** include a DNA sample, DNA primers, free nucleotides called ddNTPs, and DNA polymerase. The various components**required** for **PCR** include a DNA sample, DNA primers, free nucleotides called ddNTPs, and DNA polymerase.

9. Sheep cloned through nuclear transfer

"**Sheep Cloned by Nuclear Transfer** from a Cultured Cell Line" (1996), **by** Keith Campbell, Jim McWhir, William Ritchie, and Ian Wilmut. ... This experiment led the Roslin team to later **clone** mammals from adult body cells and to genetically engineer mammals

10. Antibodies on surface of B cells

IgM, IgD

11. Signalling in olfaction

An important contributing factor for the high sensitivity of sensory systems is the exquisite sensitivity of the sensory receptor cells. We report here the **signaling** threshold of the **olfactory**receptor neuron (ORN).

12. 3 Codon

A series of **codons** in part of a messenger RNA (mRNA) molecule. Each **codon** consists of **three** nucleotides, usually corresponding to a single amino acid.

13. peptide bond

A **peptide bond** is an amide type of covalent chemical **bond** linking two consecutive alpha-amino acids from C1 (carbon number one) of one alpha-amino acid and N2 (nitrogen number two) of another along a**peptide** or protein chain.

14. T cells B cells difference

An important **difference** between **T-cells** and **B-cells**is that **B-cells** can connect to intact antigens right on the surface of the invading virus or bacteria. This is**different** from **T-cells**, which can only connect to parts of virus antigens on the outside of infected **cells**.

15. adult stem cells benefits .

The stem cells can be found in many different regions of the body; bone marrow, the brain, skin tissue, **fat** tissue, hair follicles, and many other locations. Some potential advantages of adult stem cells: They have been proven to be effective.

People who might benefit from stem cell therapies include those with spinal cord injuries, type I **diabetes**, Parkinson's disease, Alzheimer's disease, **heart disease**,**stroke**, burns, **cancer** and **osteoarthritis**

16. Function of centromere in cell division

Centromere Functions. One major **function** of a**centromere** is joining the sister chromatids. ... On each chromatid, the kinetochore forms at the**centromere** region of the DNA. Once all of the chromatids are attached to the mitotic spindle, the microtubules pull the sister chromatids apart into the two future daughter **cells**

17. Missense mutation

In genetics, a **missense mutation** is a point **mutation**in which a single nucleotide change results in a codon that codes for a different amino acid. It is a type of nonsynonymous substitution.

18. Function of n terminal singnaling of ER

The **N-terminus** is the start of a protein or polypeptide referring to the free amine group (-NH2) ... 1 Chemistry; 2 **Function** ... The **N-terminal signal** peptide is recognized by the **signal** recognition particle (SRP) and results in the targeting ... In eukaryotic cells, these proteins are synthesized at the rough **endoplasmic reticulum**.

19. Gel electrophoresis

Gel electrophoresis is a method for separation and analysis of macromolecules and their fragments, based on their size and charge.

20. What is meant by polarity of water??

Water is a "**polar**" molecule, meaning that there is an uneven distribution of electron density. **Water** has a partial negative charge near the oxygen atom due the unshared pairs of electrons, and partial positive charges near the hydrogen atoms.

21. Steps involved in E.coli signaling
??????????

22. Examples of RNA viruses

This nucleic acid is usually single-stranded RNA (ssRNA) but may be double-stranded RNA (dsRNA).**Notable human** diseases caused by RNA viruses include Ebola hemorrhoragic fever, SARS, Rabies,**common** cold, influenza, hepatitis C, West Nile fever, polio and measles.

23. Cloning

Cloning is the process of producing genetically identical individuals of an organism either naturally or artificially. In nature, many organisms produce clones through asexual reproduction

24. paracrine signaling

Paracrine signaling is a form of cell-to-cell communication in which a cell produces a signal to induce changes in nearby cells, altering the behavior of those cells.

25. down syndrome

a congenital disorder arising from a chromosome defect, causing intellectual impairment and physical abnormalities including short stature and a broad facial profile. It arises from a defect involving chromosome 21, usually an extra copy (trisomy-21).

26. regulatory regions

Regulatory sequence. A **regulatory** sequence is a segment of a nucleic acid molecule which is capable of increasing or decreasing the expression of specific genes within an organism. Regulation of gene expression is an essential feature of all living organisms and viruses.

27. Cell cycle

The cell cycle is a cycle of stages that cells pass through to allow them to divide and produce new cells. It is sometimes referred to as the "cell division cycle" for that reason.New cells are born through the division of their "parent" cell, producing two "daughter" cells from one single "parent" cell.Daughter cells start life small, containing only half of the parent cell's cytoplasm and only one copy of the DNA that is the cell's "blueprint" or "source code" for survival. In order to divide and produce "daughter cells" of their own, the newborn cells must grow and produce more copies of vital cellular machinery – including their DNA.The two main parts of the cell cycle are mitosis and interphase.Mitosis is the phase of cell division, during which a "parent cell" divides to create two "daughter cells."The longest part of the cell cycle is called "interphase" – the phase of growth and DNA replication between mitotic cell divisions.Both mitosis and interphase are divided into smaller sub-phases which need to be executed in order for cell division, growth, and development to proceed smoothly. Here we will focus on interphase, as the phases of mitosis have been covered in our "Mitosis" article.Interphase consists of at least three distinct stages during which the cell grows, produces new organelles, replicates its

DNA, and finally divides.Only after the cell has grown by absorbing nutrients, and copied its DNA and other essential cellular machinery, can this "daughter cell" divide, becoming "parent" to two "daughter cells" of its own.The graphic below shows a visual representation of the cell cycle. The small section labeled "M" represents mitosis, while interphase is shown subdivided into its major components: the G_1, S, and G_2 phases